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# AI Builder Cup

Build in JAPAC. Build for the world.

## Kirana Mitra — a voice-first AI mentor for India’s corner shops

## Team Details

a. Team name: **singapuria_innovators**
b. Team leader name: **Vandana Soni**
c. Problem Statement: **Retail & Commerce — Intelligent Customer and Business Experiences**

# Brief about the idea

**Ramesh Gupta, 44 · Gupta General Store, Indore · 200 sq ft, 14-hour days, a ₹9,000 Android phone is his only computer**

Pain point #1: he finds out Maggi is finished when a customer asks for it. Ordering is a phone call from memory, so he over-buys slow items and runs out of fast ones. Every stock-keeping app expects him to type, usually in English.

**Kirana Mitra** is a voice- and photo-first AI mentor for kiranas. He photographs a shelf or says what arrived, in Hindi, Hinglish or any of India's 22 scheduled languages. Gemini turns it into stock counts, a festival-aware demand forecast and a ready order for his distributor — no typing, no English. Every evening it tells him what sold, what is running low and what to order tomorrow.

*Scalable (serverless, scales to zero) · High impact (lost sales and dead stock are the two biggest leaks in a kirana) · Easy (two buttons: speak, snap)*

**~13M**

kirana stores in India — the total addressable market (Invest India)

**~91%**

of India's grocery retail still flows through them (Redseer, 2025)

**~6M**

serviceable: smartphone-owning kiranas that order from a distributor (our estimate)

Sources: Invest India (store count); Redseer via 36Kr (2025 grocery share); serviceable figure is the team's estimate

# Opportunities

### How is it different?

Most retail AI is built for chains and e-commerce. Ledger and billing apps for small shops still expect typing, often in English. Kirana Mitra needs only a phone camera and a voice, and answers in the shopkeeper's own language.

### How does it solve the problem?

It turns shelf photos and voice notes into stock records automatically, forecasts demand around India's festival calendar, and prompts the shopkeeper to place the order with the distributor before the shelf runs empty.

### Why now, and how big

13M stores, ₹200 average bill, smartphones in most shops. Gemini's native audio and vision make zero-typing input practical for the first time, and serverless GCP keeps cost per shop in paise.

## USP

**Speaks their language**

Hindi, Hinglish and the 22 scheduled languages, code-mixing included

**Zero data entry**

Photo and voice in, stock and orders out, with one-tap undo

**Built around India's calendar**

Demand forecast knows Diwali, Eid, Navratri and wedding season

# List of features offered by the solution

### Shelf check

One photo; Gemini counts packs, flags what is low and asks “Update karoon?” before anything changes.

### Voice ordering & stock entry

Say “Kal Parle-G ke 3 carton aaye”; stock updates and the next distributor order is one tap away.

### Demand & supply forecast

BigQuery ML ARIMA_PLUS with Indian holidays: 4 weeks ahead per item, Diwali bump included.

### Smart restock

Top sellers that are low and have run hot for 2 weeks get an order drafted and a WhatsApp nudge to the distributor.

### 22-language mentor

Replies in the shopkeeper’s language and script; mixed Hinglish stays Hinglish.

### Evening hisaab & feedback

What sold, what is low, what to order tomorrow — and a thumbs-up or down that reaches the team.

# Process flow and use cases

1
**Shopkeeper**
Shelf photo, voice note or a typed line, in any language

2
**Cloud Run API**
FastAPI service; one prompt per task with a JSON response schema

3
**Gemini 3.8 Flash**
Vertex AI; reads photos, audio and Indic text with the catalogue in context

4
**BigQuery ML**
Sales history; ARIMA_PLUS forecast with Indian holidays

5
**Reply**
Stock update, forecast or order draft, in their language, with Undo

## A shopkeeper’s day

**9 am · Morning**
Shelf check → “Maggi 6 packet, kam hai (reorder 12). Update karoon?” → Haan

**1 pm · Daytime**
Voice note “Kal Parle-G ke 3 carton aaye” → “Samjha: 3 carton = 72 packet, ab 114”

**9 pm · Evening**
Hisaab: aaj ki bikri, kam stock, and “Diwali week demand ~1.8x — Parle-G 264 packet mangao?” → Order banao

# Wireframes of the proposed solution

## Morning: shelf check




## Daytime: voice ordering

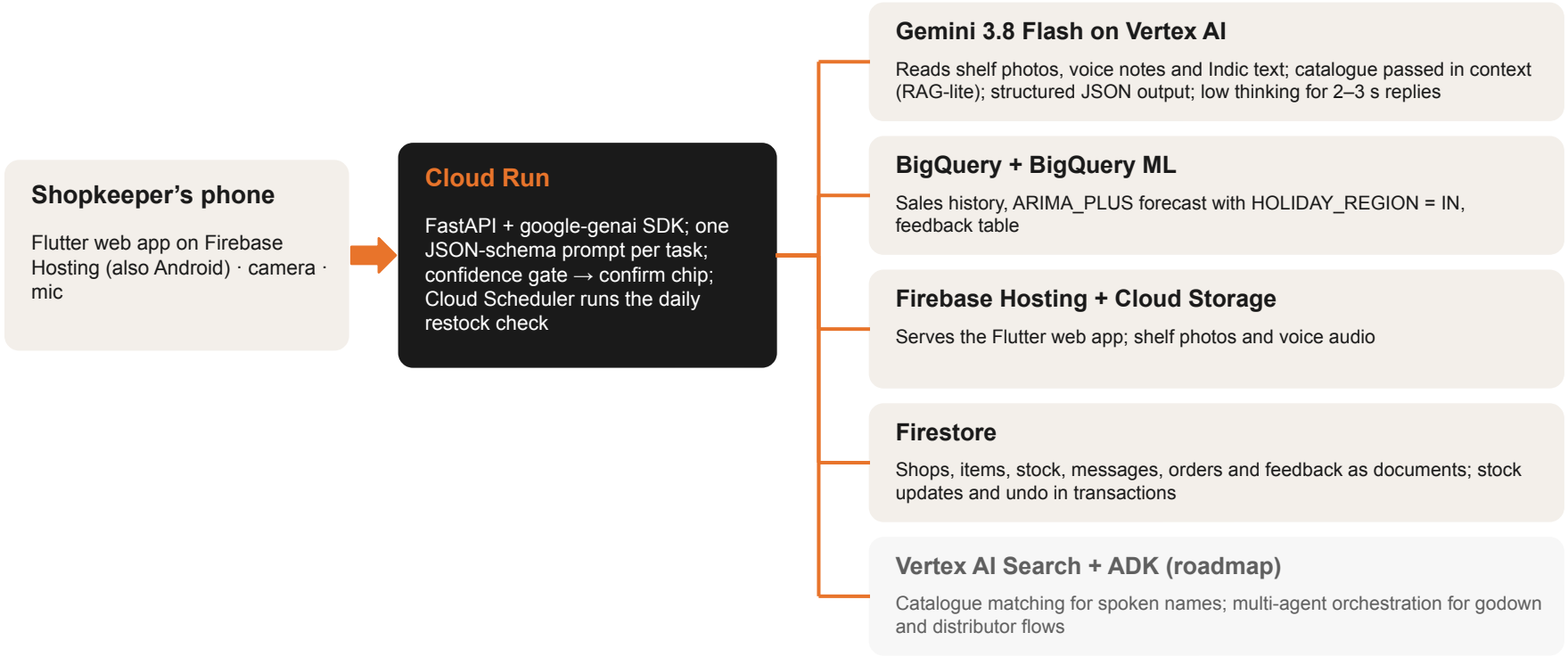



## Evening: daily hisaab




Reply strings are live output of the prototype (gemini-3.8-flash, 18 Oct 2026); figures are the demo shop's

# Architecture diagram of the proposed solution



# Technologies to be used in the solution

| Layer | Technology | What it does here |
|---|---|---|
| **AI model** | Gemini 3.8 Flash on Vertex AI, google-genai Python SDK | Reads shelf photos, voice notes and Hindi, Hinglish or regional text; replies in the same language. Structured JSON output on every call |
| **Prompting** | Few-shot + response schema + confidence gate | Hindi/Hinglish/English examples in one system prompt; catalogue in context; low thinking budget; entries under 0.7 confidence wait for a tap |
| **Analytics** | BigQuery, BigQuery ML ARIMA_PLUS | Sales history and 4-week demand forecast with Indian holidays; festival uplift; feedback table |
| **Backend** | Cloud Run + Cloud Scheduler | FastAPI API; scales to zero when idle; daily auto-restock check with a token-gated endpoint |
| **Database** | Firestore (Native mode, asia-south1) | Shops, items, stock, messages, orders and feedback; transactional stock updates and undo |
| **Hosting / storage** | Firebase Hosting, Cloud Storage | Serves the Flutter web app; photos and audio |
| **Frontend** | Flutter (web now, Android APK built) | Chat-style app: hold-to-talk, camera, 56 dp targets, works on low-end phones |
| **Roadmap** | Firebase Auth · Vertex AI Search · ADK · on-device model | Sign-in, catalogue matching, multi-agent orchestration, offline image recognition |

# Estimated implementation cost

Pilot of 100 shops in Indore, per month. Assumes 20 voice or text entries and 2 shelf photos per shop per day.

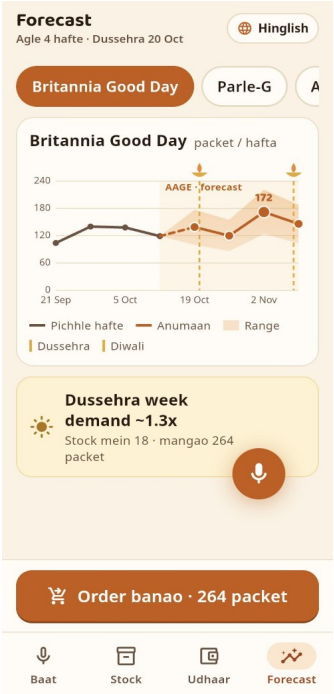
| Component | What drives the cost | Est. per month |
|---|---|---|
| **Gemini 3.8 Flash on Vertex AI** | ≈0.9M tokens per shop per month (prompt + catalogue, audio and image input; short JSON output) | ≈ ₹8,500 (≈ ₹85 per shop) |
| **Cloud Run + Cloud Scheduler** | ≈70k requests per month, scales to zero; scheduler job is in the free tier | ≈ ₹0 (free tier) |
| **BigQuery + BigQuery ML** | Daily ARIMA_PLUS retrain measured at ₹0.25 per run; queries inside the 1 TB free tier | ≈ ₹10 |
| **Cloud Storage** | Web app, photos and audio, a few GB | ≈ ₹20 |
| **Firestore** | Documents per shop; inside the free tier at pilot volume | ≈ ₹0 (free tier) |
| **Total today** | ≈ ₹85 per shop per month at Gemini 3.8 Flash introductory pricing to 31 Dec 2026; about double from 2027 | **≈ ₹8,550** |

Pricing: Gemini 3.8 Flash $0.75 / $3.75 per 1M input / output tokens (introductory), ₹84 per $. Measured items: BQML training ₹0.25 per run and evaluation ₹0.05 (bytes billed, 18 Oct 2026). Gemini figure from the Vertex AI pricing page for Gemini 3.8 Flash at the assumed volume; rerun with the pricing calculator before the pilot. Reduce-cost roadmap: on-device image model for shelf checks, Gemini Flash-Lite for simple intents, caching the catalogue prompt.

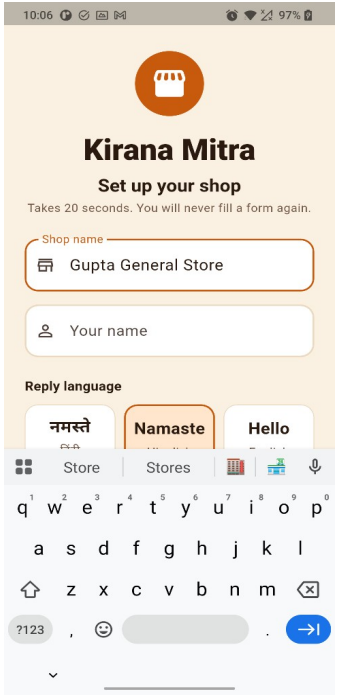
# Snapshots of the prototype




Voice or typed entry → confirmation card with Undo (live, Cloud Run)




4-week forecast with Dussehra and Diwali markers, reorder in one tap




The same app on a ₹10k Android phone (OnePlus, Android 11)

Kirana Mitra · singapuria_innovators

# Prototype performance report / benchmarking

| Metric | Target | Result |
|---|---|---|
| **Shelf photo item-count accuracy** | ≥ 90% | **96.2% exact counts (25 of 26), recall 100%** |
| **Voice / text entry accuracy (Hindi, Hinglish, English)** | ≥ 90% | **96.2% intent · 100% entities (26 utterances)** |
| **Forecast error, 4 weeks ahead** | ≤ 20% | **11.0% volume-weighted (19.6% without festival uplift)** |
| **Time from photo to reply** | < 5 s | **9.0 s median (text entry: 2.5 s median)** |
| **Cloud cost per shop per month** | < ₹100 | **≈ ₹85 (introductory Gemini pricing; measured BQML ₹0.25 per retrain)** |

Test set (18 Oct 2026): 5 synthetic shelf images with 26 labelled product counts; 26 Hindi, Hinglish and English utterances; forecast back-tested on a 14-day holdout of 120 days of synthetic sales for 40 items (Navratri inside the window). Photo latency misses target because of Gemini tail latency; a 40 s cap with one retry keeps the app responsive. Real-shelf photos and real sales are the next measurements.

# Additional details / future development

### Godown plugin

Warehouse stock alongside the shelf: shelf check → godown double-check → then prompt the kirana to place the order with the distributor.

### Offline image model

On-device shelf recognition (Gemini Nano or a TFLite model) so counts work without network and Gemini cost drops for the most frequent call.

### Production gaps to close

Sign-in and per-shop access (Firebase Auth) · real pilot sales and shelves · monitoring, rate limits and Gemini cost caps · backups and Firestore security rules · native review of the 22 languages.

### 22 languages in the app

Backend already replies in all 22; the generated UI strings ship language by language after native-speaker review, largest eight first.

### Udhaar ledger & credit

Voice-logged credit and reminders; clean sales history can help shops qualify for working-capital loans.

### Supply forecast & WhatsApp

Distributor lead times and stock-outs in the forecast; WhatsApp-first entry so shopkeepers never install anything.

*Business model: free for shopkeepers; distributors and FMCG brands pay for anonymised, aggregated demand insights and for the restock channel.*

# Provide links to your:

**GitHub public repository**

github.com/vipindev-shambu/kirana-mitra

**Demo video (3 minutes)**

[video link]

**Final product**

storage.googleapis.com/kirana-mitra-app-web/index.html · API: kirana-mitra-api-396621755403.asia-south1.run.app

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# Thank you

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